

vider. Sensors use a reaction-based approach to manage and maintain an asset and maximise its use potential. A wide range of sensors is available to collect huge amounts of data from all possible systems of a single train and then analyse it in real time to detect problems before these actually occur.

Constant monitoring of equipment through the measuring of all relevant variables such as temperature, vibrations, oil levels and the like help anticipate the optimal timing for maintenance. It enables identification of faults proactively and elimination of necessary maintenance interventions. Predictive maintenance is a powerful tool that helps track asset health, reduce unplanned downtime of equipment and minimise the high cost of unscheduled maintenance.

Optical and tactile sensors such as light curtains, camera systems and dynamic pressure-sensitive mats are suited to monitoring areas near rail vehicles. Good internal communication, fast reactions based on equipment geolocation data, high-quality maintenance planning and regular interventions are required to keep massive rail networks working.

Predictive maintenance and CMMS for improved railway asset performance

Modern, next-generation asset and maintenance management starts with the adoption of a smart computerised maintenance management system (CMMS). Reliable railway maintenance is required to improve critical issues like safety, delays and overall system capacity. It is expected to rely on smart transportation systems and interconnected solutions such as predictive maintenance.

An interconnected CMMS can help maintain, manage and connect tracks, terminals, rolling stocks and communications infrastructure. It can identify maintenance issues before these impact safety, operations or revenue. It collects, stores and analyses data to prevent breakdowns and issue predictive maintenance algorithms to extend equipment life.

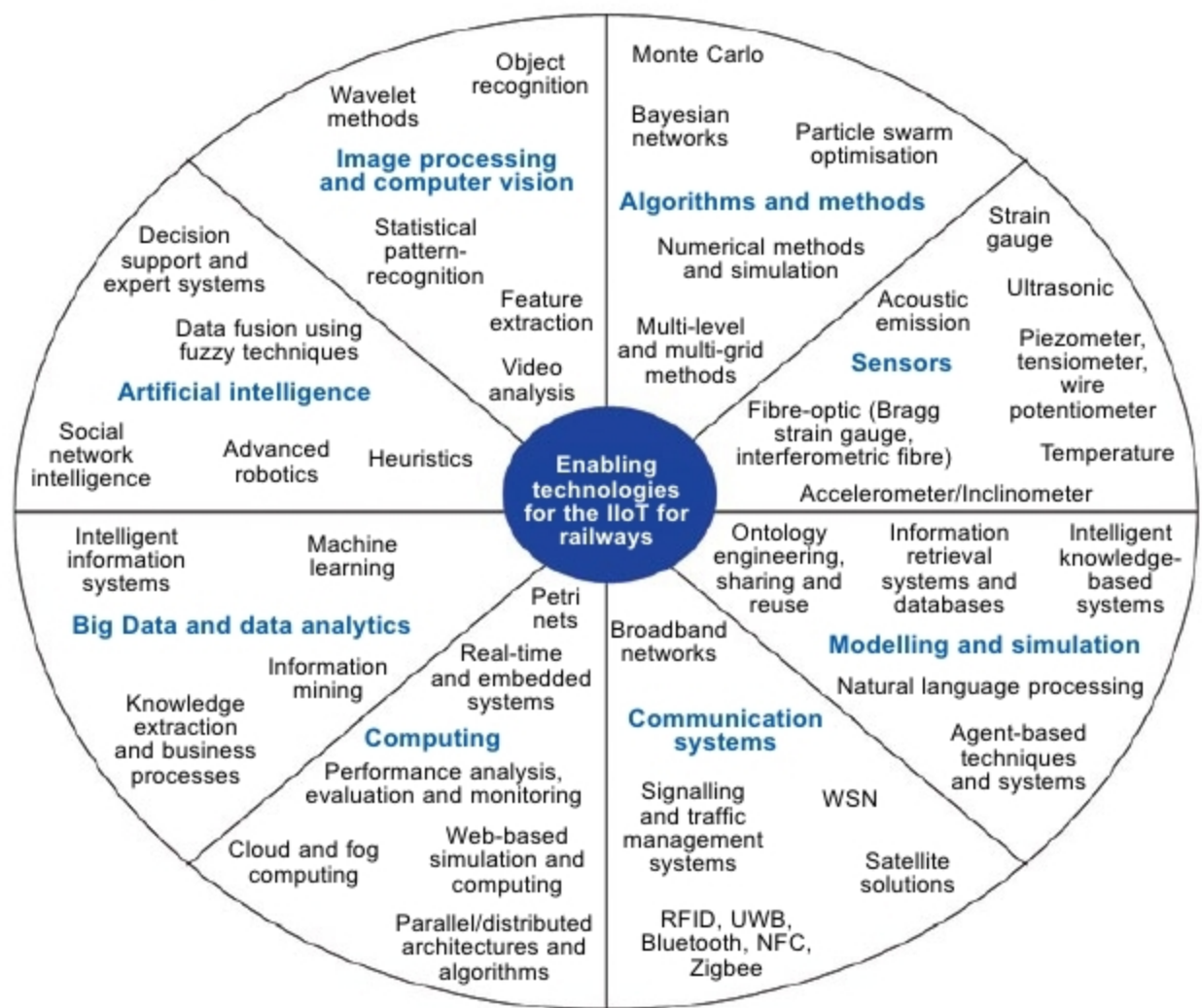


Fig. 2: Enabling technologies for the IIoT of railways (Credit: www.mdpi.com)

A reliable CMMS should be user-friendly, fast, reactive and flexible. It should also have a mobile application for anytime access, connectable to ERPs and IoT systems, geolocation tool, an analytical tool that supports unrestricted media upload and so on. Different departments such as accounting, operations, purchase and maintenance should also be connected to the entire communication platform. It can work as a network for manufacturers, technicians and suppliers to exchange expertise and speed up operations. Advanced CMMS analytic tools enable organisations to analyse data with great speed and accuracy, to optimise availability and increase the life of assets.

The IoT-connected trains

The IoT can interconnect all objects and devices that were previously not part of a network for predictive analytics. Its application increases safety, efficiency and ease of use with train management systems. Control and surveillance systems reduce the risk of collisions and regulate speed. Advanced consumer technologies help maximise connectivity and

allow passengers to continue their activities on smart devices while travelling. Train-to-train communication through the cloud enables operators to transmit data about equipment, tracks and stations among themselves.

The IoT enables monitoring of areas on railway crossings remotely, such as barrier operations and end positions, switch end positions, space between barriers, system operations, connections and signals. This allows users to accelerate their projects, from engineering and runtime to maintenance with fast detection and localisation of errors and faults.

Big Data analytics for smart railways

The complete Big Data architecture includes the IoT and cloud computing devices. These work together to create smart railways that have self-learning capabilities to predict failure, make diagnoses and trigger maintenance actions. The architecture utilises multiple data sources to extract relevant information. It helps users to know what happened when, so they can go back and do the root cause analysis from the data, and take appropriate corrective action.